

Basic Principles of Medicine 1
Module : Foundations to Medicine
DLA

Early Embryology Week 1

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Recommended Reading

Moore KL, Persaud TVN, Torchia MG. The Developing Human – Clinically Oriented Embryology, 11th Ed. Chapters 1-4. pgs. 37-39, 47 - 56

- Pay special attention to the Green boxes for the clinical correlations for this topic.

Objectives

SOM.MKII.BPM1.1.FTM.1.ANAT.1001	Briefly describe the process from fertilization to the formation of the blastocyst, paying attention to the location and timeframe of each event
SOM.MKII.BPM1.1.FTM.1.ANAT.1002	Describe the location and process of implantation of the blastocyst
SOM.MKII.BPM1.1.FTM.1.ANAT.1003	List the derivatives of the germ layers.

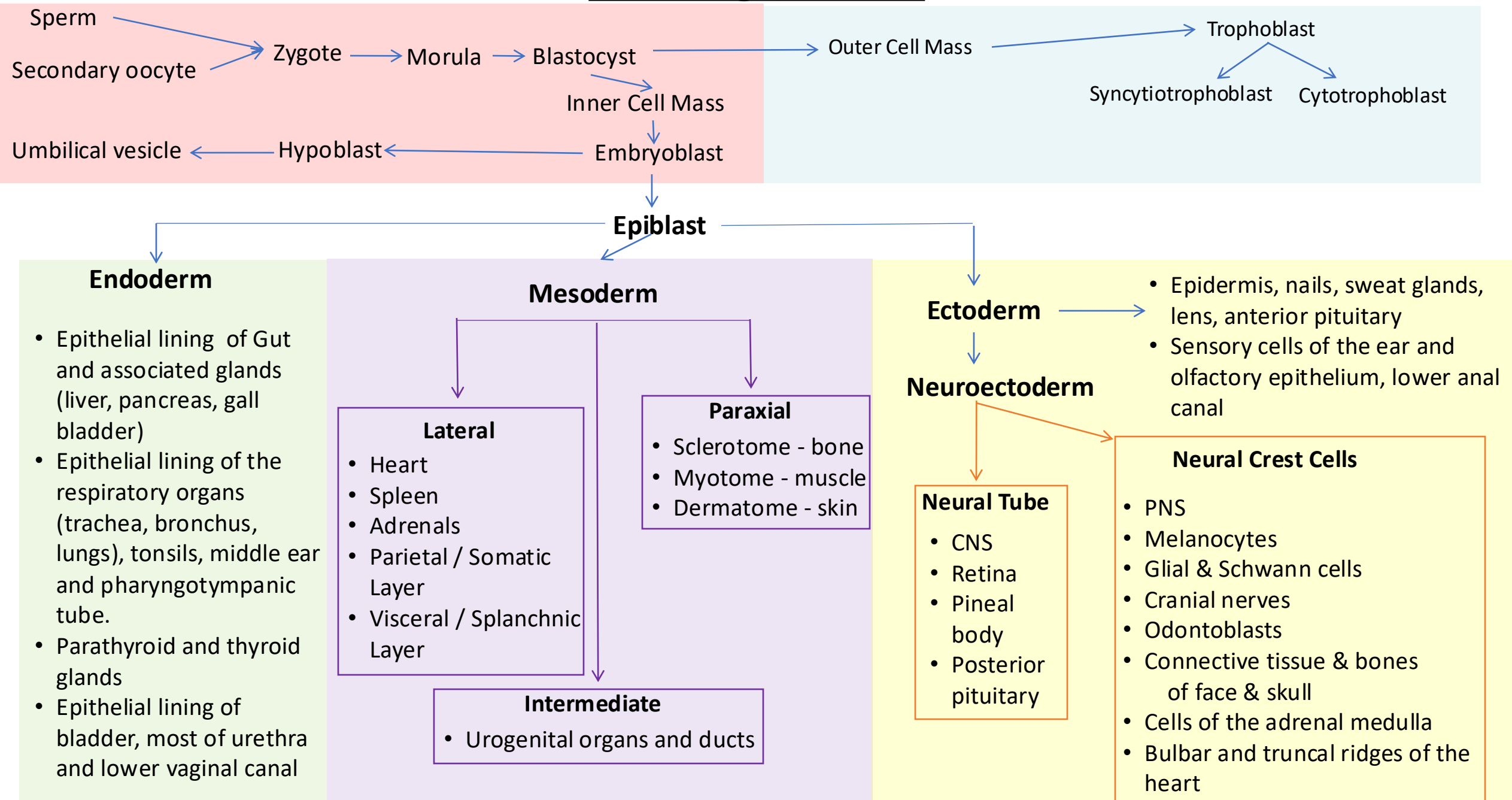
Special note

Some slides have a green background, this will not be covered in the video recording but are put so you have the information available

Terminology

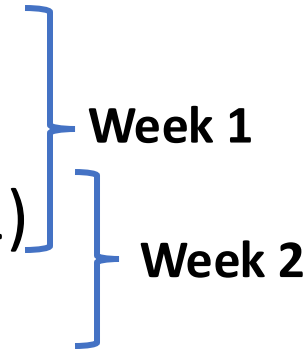
- **Cytokines** – small proteins produced by cells that act as messengers for other cells
- **Coelom** – Space/cavity
- **Syncytium** – multinucleated cell mass
- **Chemotaxis** – movement of cells toward or away from a chemical signal
- **Blast** - cell (usually immature form)
- **Cleavage** – rapid mitotic division of cells
- **Differentiation** - process whereby cells acquire their mature morphological and biochemical characteristics. Differentiation is often considered a 'final step' of development in which cells take on their mature function.
- **induction** - alteration in cell fate resulting from external signal (inducer), a form of cell-cell signaling
- **ingression** - migration of individual cells from surface layers into the interior of the embryo
- **involution** - a type of cell movement during gastrulation which involves the inturning or inward movement of an expanding outer layer so that it spreads over the internal surface of the remaining external cells
- **mesenchyme** - mesodermal cells in a developing embryo with the ability to move freely and individually

The Big picture

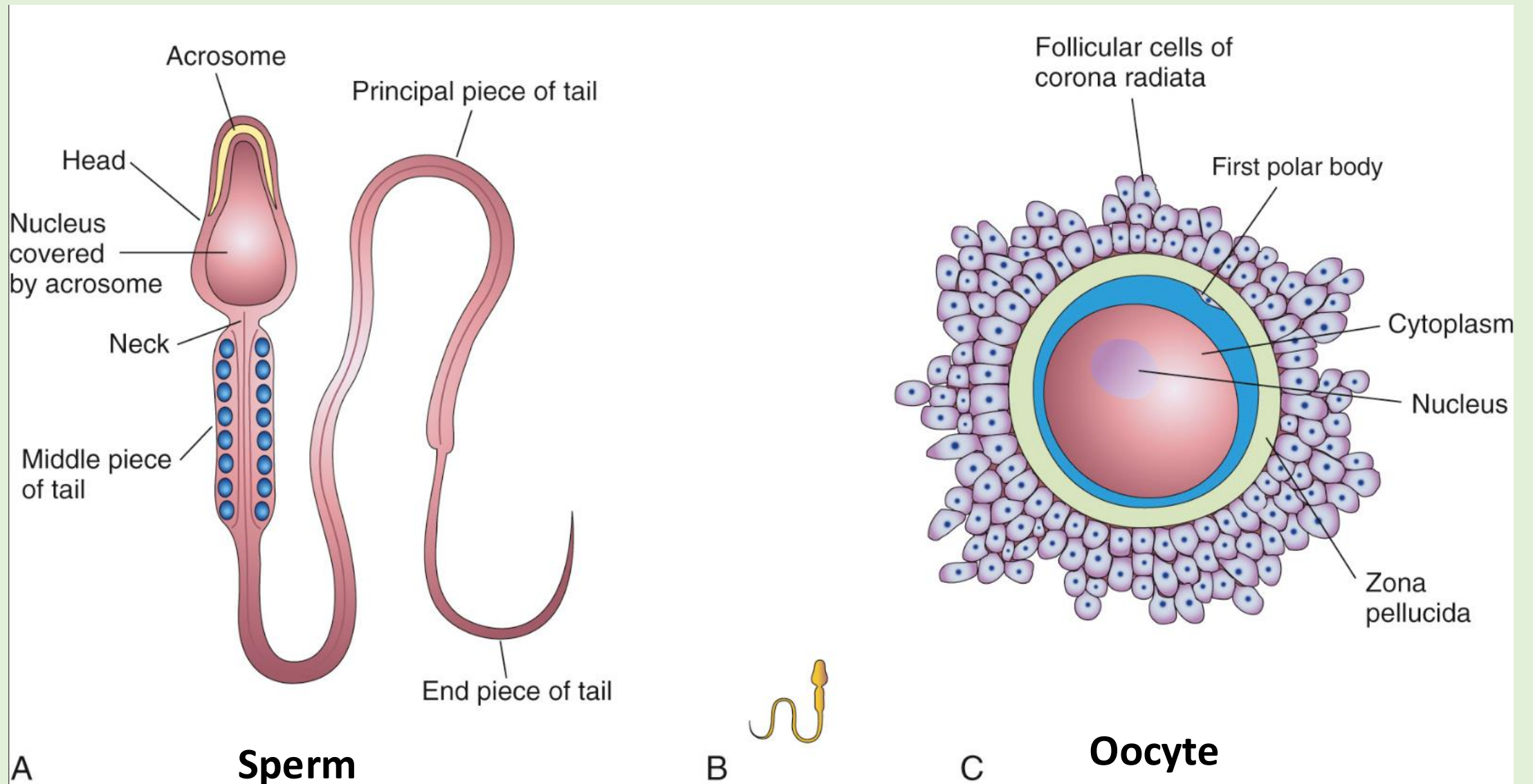


The journey

- The development of the embryo from male and female gametes to a fully formed embryo is termed Embryogenesis
- It consists of a series of events
 - Fertilization – Day 1
 - Cleavage – Day 1 - 6
 - Implantation – Day 6 – 10 (beginning of week 2)
 - Formation of the bilaminar disc – Day 10 - 14
 - Gastrulation (formation of embryonic cell lines) - Day 14 – 21 (**Week 3**)
- Organogenesis – **Week 4 to 8**
- Everything happens synchronously with many simultaneous events



Gamete structure

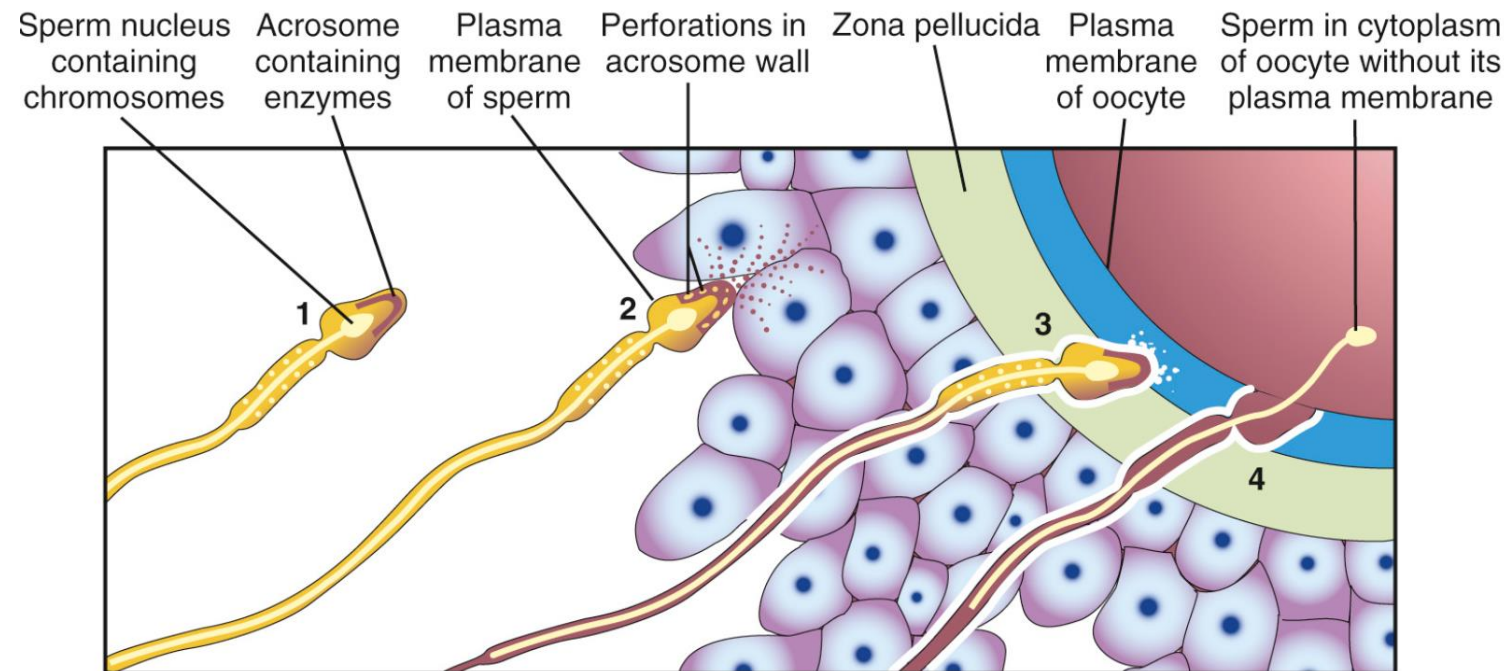
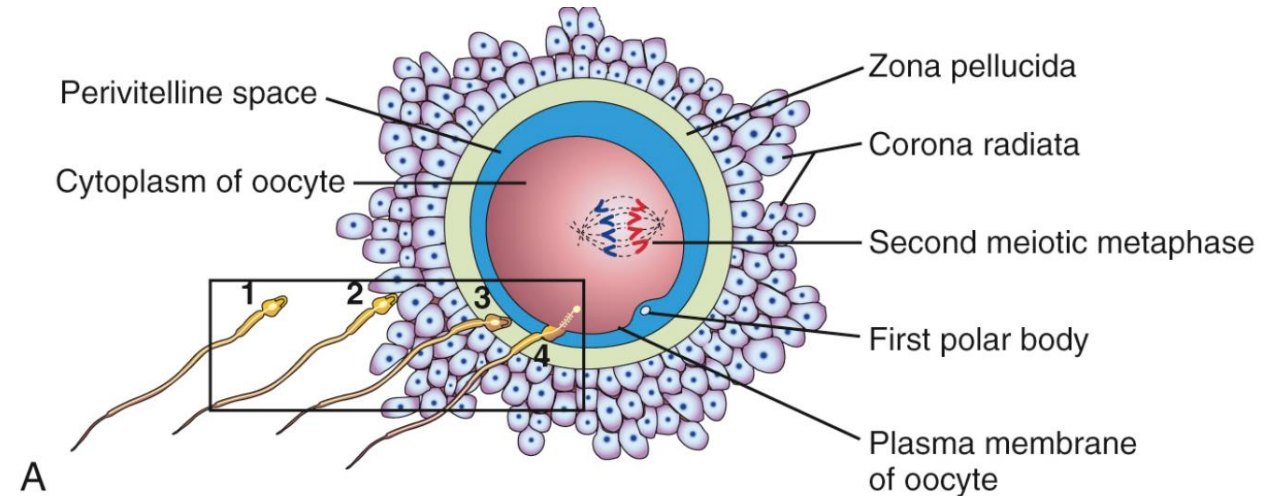


Week 1, Day 1 – Fertilization

- Where? **Ampulla** of the uterine tube
- When? **12 and 24 hours** after ovulation
- How? **Chemotaxis**

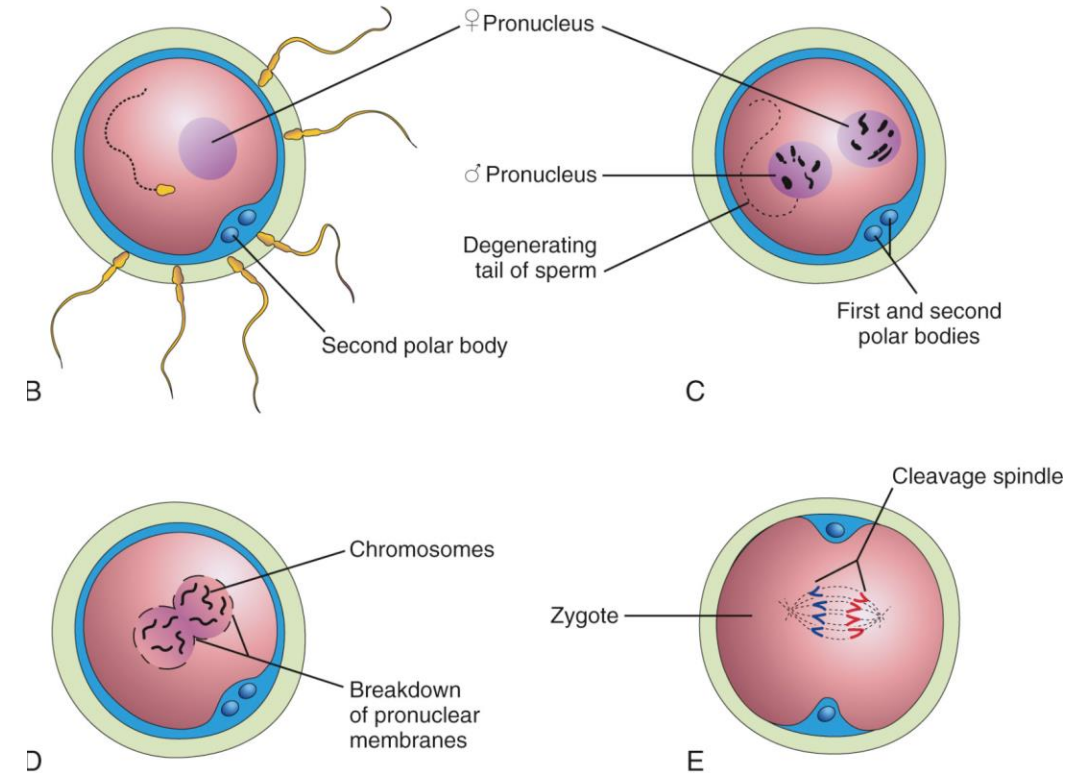
Sequence of events

- Capacitation (2)
- Acrosome reaction (3)
- Fusion of the plasma membranes of sperm and oocyte (4)
- Cortical reaction – prevents other sperm from entering
- Sperm chromatin decondensation to form male pronucleus:
- Oocyte nucleus completes the second meiosis



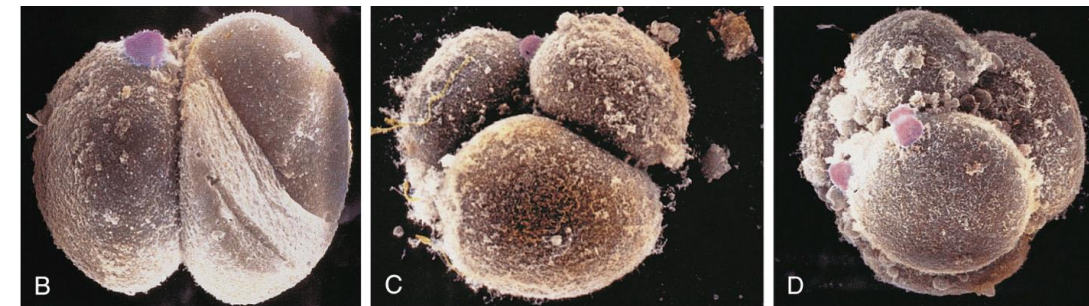
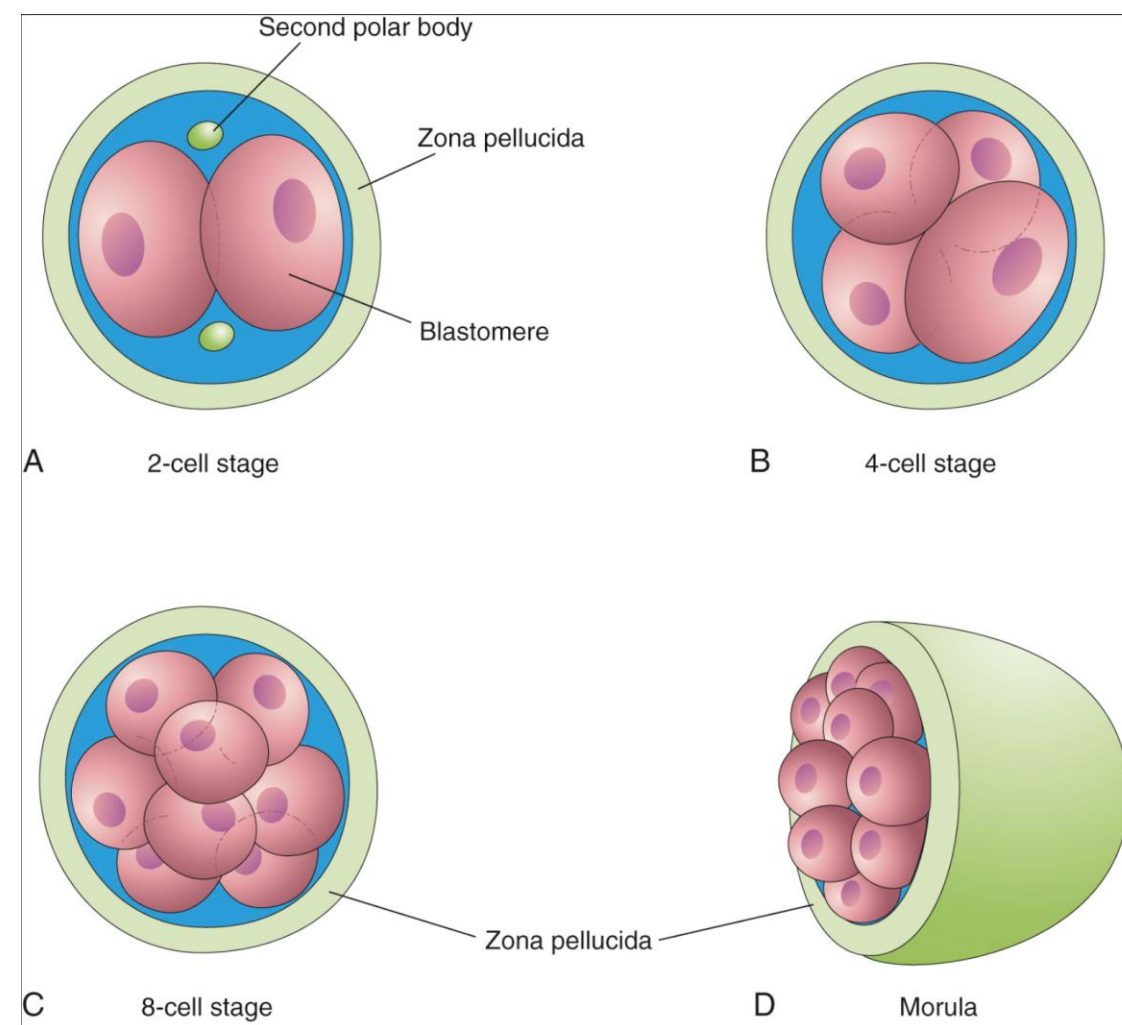
Week 1, Day 1 – Fertilization cont.

- **Fusion of pronuclei:**
 - Pronucleus of sperm fuses with pronucleus of oocyte
 - **Restores the normal diploid number of chromosomes**
 - **Zygote** is formed
 - Embryonic sex is determined
- DNA replication
- Mitotic division and **cleavage begins**



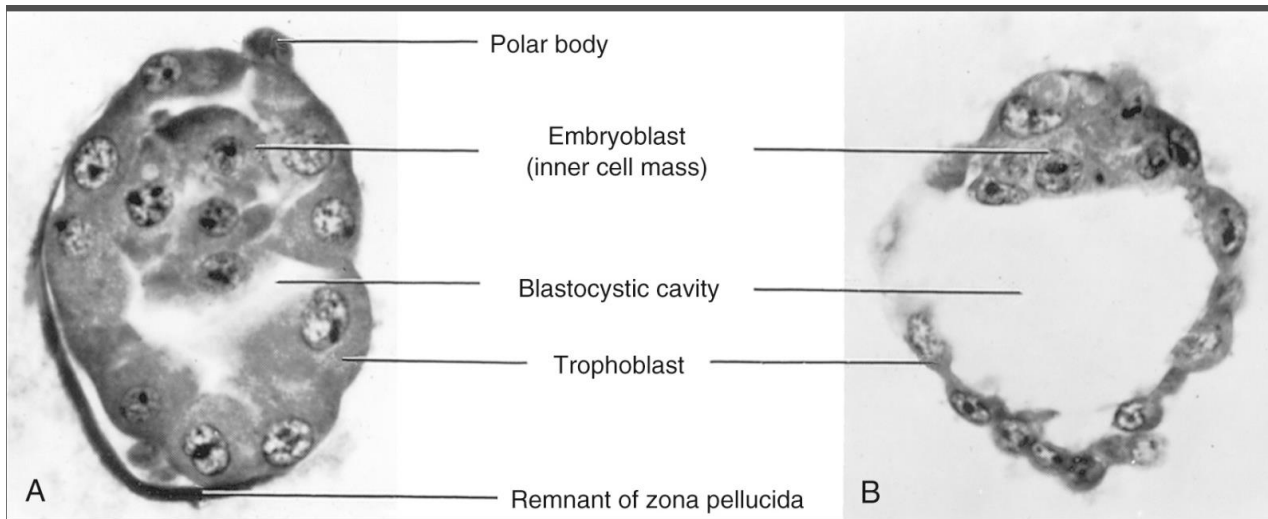
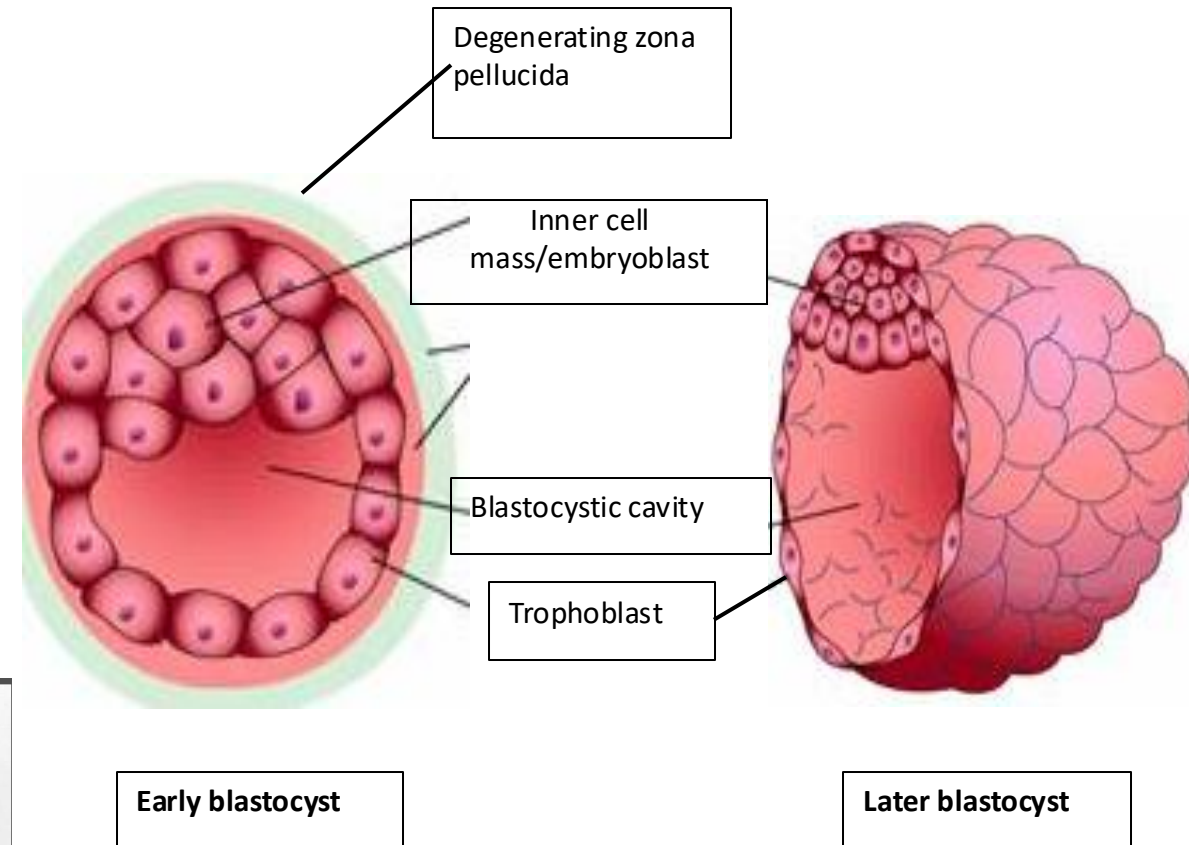
Week 1, Day 2 - Cleavage

- ~ 30 hours after fertilization
- Repeated mitotic division of the cells (**blastomeres**) of the zygote
- Surrounded by **zona pellucida**
- Moves through the **uterine tube** toward the uterine cavity
- At 9 cell stage cells change shape and forms a compact ball – compaction
 - **Segregation of cells** forming the embryo and the membranes
 - Rostro-caudal and lateral **axes** is established
- **Morula (Day 3)** 12 - 32 blastomeres
 - distinction between the inner mass (**embryo**) and the trophoblast (**membrane forming**)



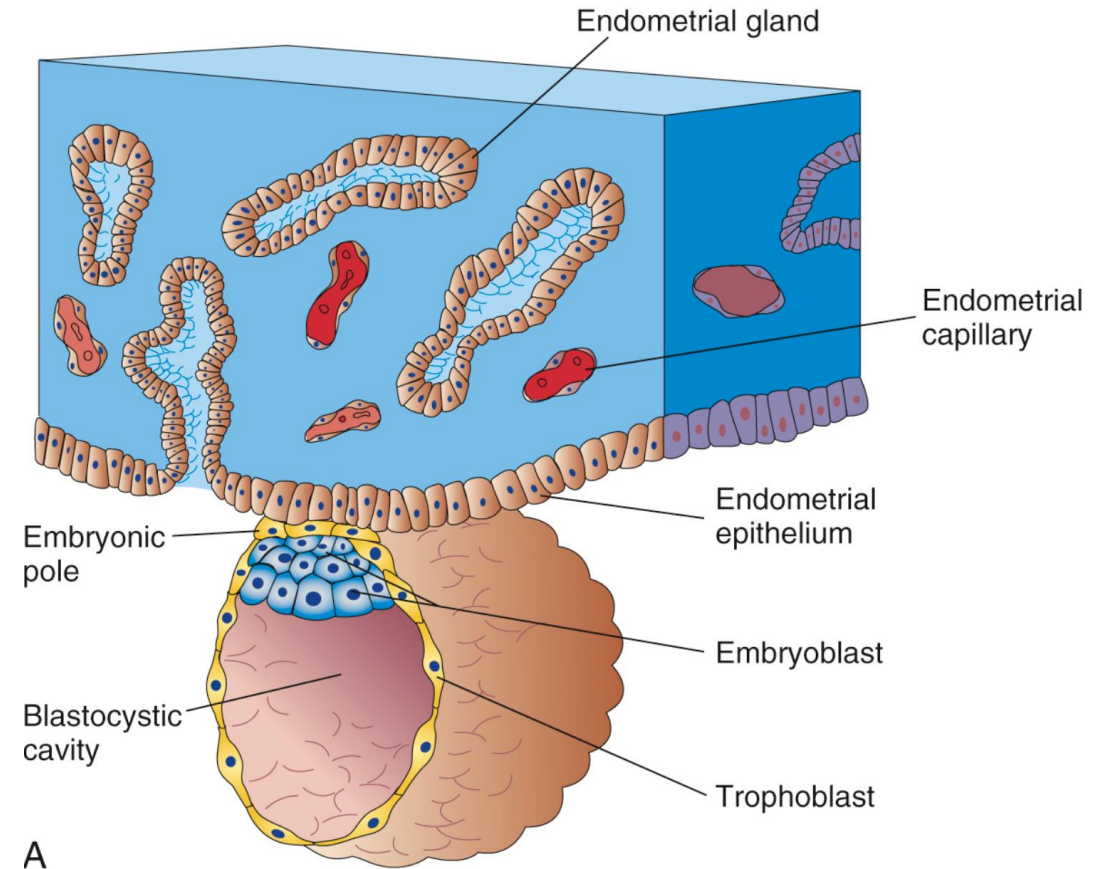
Week 1, Day 4 - Blastocyst Formation

- Morula enters the uterine cavity
- **Fluid** from the uterine cavity enters the morula
 - Blastocystic cavity
 - Separates the blastomeres into:
 - **embryoblast** (future embryo)
 - **trophoblast** (fetal component of the placenta)
- Zona pellucida starts to degenerate



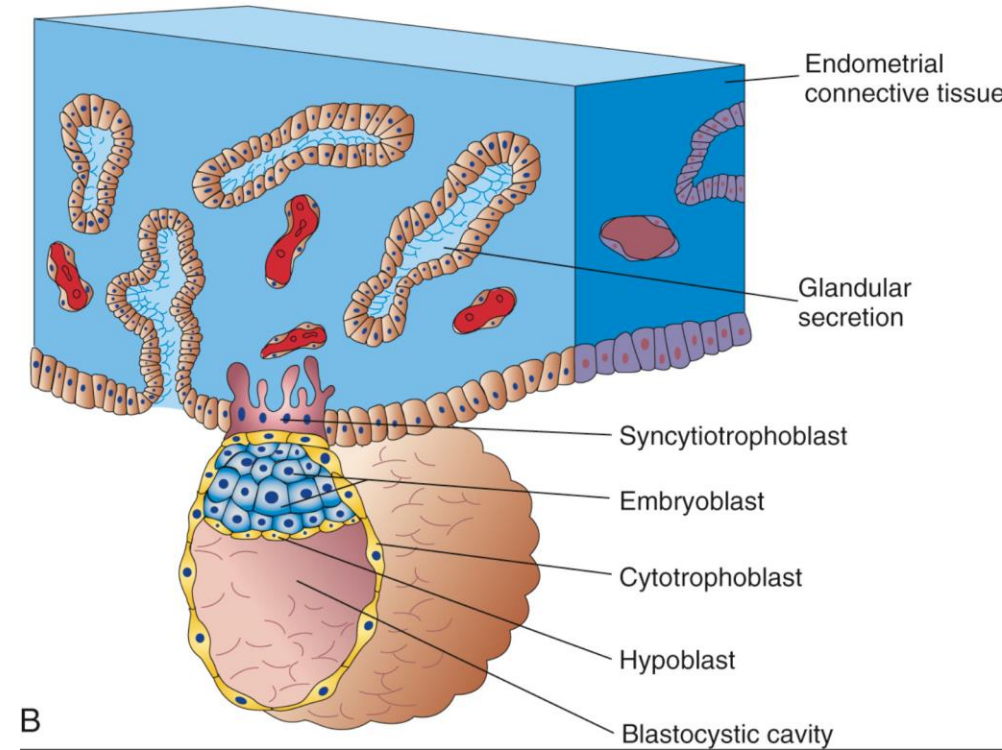
Week 1, Day 6 – Implantation starts

- Starts at day 6 and continues till day 10
- Blastocyst attaches to the epithelium of the endometrium (inner lining of the uterus),
 - **Posterior wall of body of uterus close to the fundus**
- Trophoblast proliferates rapidly and differentiates into
 - An outer layer of **syncytiotrophoblast**
 - An inner layer of **cytotrophoblast**

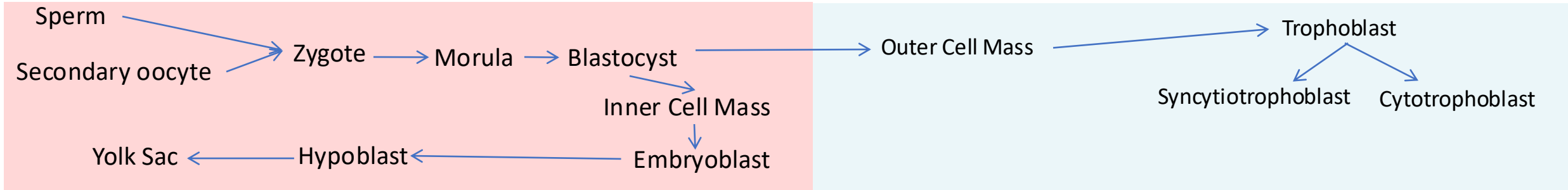


Implantation cont.

- Outer **Syncytiotrophoblast** – **S**ecretes and **I**nvades
 - Multinucleated mass
 - Fingerlike projections **invade** into the connective tissue
 - **Secretes enzymes** that erode maternal tissue
 - secrete human chorionic gonadotrophin (**HCG**)
- Inner **Cytotrophoblast** – makes **C**ells
 - Mitotically active
 - Forms new cytotrophoblast
- **Day 7** – formation of **hypoblast** on the surface of the embryonic pole that faces the blastocystic cavity
 - Marks the start of changes in the embryoblast



Summary of week 1



- Fertilization – Day 1
- Cleavage – Day 1 – 6
 - Formation of morula
 - Formation of blastocyst
- Implantation – Day 6 on...
 - Division of embryoblast and trophoblast
- Development of hypoblast – Day 7

